Reflective Journal – Midterm Project: Healthcare Stroke Dataset

John’a Matthews

1. Introduction

For this project, I worked with the Healthcare Stroke Dataset to learn how to explore, clean, and prepare data for machine learning. The main purpose was to understand the dataset, identify data-quality problems, and apply preprocessing techniques that could help prepare the information for a future stroke prediction classification model. I used Python, Pandas, Matplotlib, and Scikit-learn in Google Colab to examine the data and complete the preprocessing steps. This project helped me understand that machine learning involves more than writing code or choosing a model. The quality of the data and the way it is prepared can have a major impact on the results. It also helped me connect the concepts I learned in class to a healthcare-related example.

2. Description of the Experience

I began by loading the dataset into Google Colab and examining its structure. I checked the number of rows and columns, reviewed the data types, identified missing values, checked for duplicate records, and generated descriptive statistics. This gave me a better understanding of the information available and helped me identify areas that needed attention before using the data for machine learning.

After inspecting the original dataset, I divided the data into 70% training data and 30% testing data. I learned that this step is important because the training data is used to develop a model, while the testing data is reserved for evaluating how well the model performs on data it has not seen. I also learned that preprocessing decisions should be fitted using the training data and then applied to the testing data to help prevent data leakage.

Next, I performed exploratory data analysis using summary statistics and visualizations. I examined numerical features such as age, average glucose level, and body mass index (BMI). I also explored the distribution of the stroke target variable to understand the difference between the two classes. I used Matplotlib to create charts that made it easier to see patterns and differences in the data.

I then practiced several preprocessing techniques. These included handling missing values, scaling numerical features, encoding categorical variables, and creating additional features from existing information. I also explored distribution transformation for numerical data and class balancing for the training dataset. Finally, I compared samples before and after preprocessing and prepared cleaned training and testing datasets for future machine learning work.

3. Personal Reflection and Learning

One of my biggest learning experiences was understanding why missing values need to be handled carefully. For example, BMI contains missing entries in this dataset. I learned that using the median can be a practical way to fill missing numerical values because it is less affected by unusually high or low values than the mean. I also learned that categorical data, such as work type and smoking status, needs to be represented in a numerical format before many machine learning algorithms can process it. One-hot encoding helped me understand how categories can be converted into separate features without assigning them an incorrect numerical ranking.

I also learned why class imbalance is important when preparing data for machine learning. In this dataset, there are many more records of patients who did not have a stroke than patients who did. I realized that a model could have high accuracy by mostly predicting the majority class while failing to identify enough stroke cases. By exploring oversampling, I learned how to balance the classes in the training data, so the model has a better opportunity to learn from the less common class. I also learned that the testing data should keep its original class distribution to provide a more realistic evaluation of the model's performance.

4. Conclusion

Overall, this project improved my understanding of exploratory data analysis and data preprocessing. I gained experience using Python and Google Colab to inspect the healthcare dataset, identify missing values, create visualizations, encode categorical features, scale numerical values, engineer features, and explore class balancing. I also learned the importance of splitting data correctly and fitting preprocessing steps on the training data to reduce the risk of data leakage.

My biggest takeaway is that good machine learning begins with understanding the data. I feel more confident organizing a preprocessing workflow, explaining why I use specific techniques, and checking whether the results make sense. This project also showed me that there is still more to learn about model evaluation, feature selection, and improving predictions. I plan to use these skills in future assignments as I continue learning how to build and evaluate machine learning projects.